

STATIC INVENTORY PROBLEMS UNDER RISK

1. Introduction

Static inventory problems deal with situations where only one ordering decision is taken for a fixed period. Demand is uncertain but probabilities are assumed to be known. These are also called single-period inventory problems.

2. General Characteristics

Only one procurement decision, uncertain demand, no replenishment, excess stock leads to loss, shortage leads to lost sales and goodwill loss. Objective is to minimize expected cost.

3. Christmas Tree Problem

A vendor selling Christmas trees orders stock before Christmas. Unsold trees after Christmas have little or no value. Ordering too many causes excess cost and ordering too few causes shortage cost.

4. Total Cost Matrix

A table showing total cost for each order quantity under each possible demand. Rows represent decisions and columns represent demand states.

5. Opportunity Cost Matrix

Opportunity cost or regret shows loss due to not choosing best decision for a given demand. It is used to minimize maximum regret.

6. Cost of Risk

Cost of risk is the expected loss due to uncertainty and is the difference between expected cost and cost under certainty.

7. Mathematical Formulation – Discrete Case

Expected Cost = $\sum (\text{Probability} \times \text{Cost})$. The order quantity with minimum expected cost is optimal.

8. Mathematical Formulation – Continuous Case

Expected Cost = $\int C(Q, D) f(D) dD$.

Optimal order is given by critical ratio:

$$P(D \leq Q) = C_u / (C_u + C_o)$$

9. Case Studies

Newspaper vendor, hospital blood bank, seasonal fashion retail, bakery items.

10. Conclusion

Static inventory models help balance excess and shortage costs under uncertain demand.